

Marcel Dabek

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EDUCATION

University of Connecticut, Storrs, CT

Aug 2023 – May 2027

Bachelor of Science, Mechanical Engineering | GPA: 3.75

PROFESSIONAL EXPERIENCE

Engineering Intern, CIMA Group - Pratt & Whitney | Full Time, On-site

May 2026 – Present

- Analyzed inspection data on F135 rotors flagged for defects to understand fatigue life on different failure modes
- Supported R&D on automated fiber placement (AFP) of composite fan blades, hand-laying prepreg tows, bagging and curing
- Performed cleaning and preventive maintenance on the AFP machine to keep the cell running between builds
- Simulated composite specimens under ASTM four-point bend testing in LS-DYNA to characterize flexural behavior

Software Development & Engineering Intern - Transcend Mechanics | Part Time, Remote

Feb 2026 – Present

- Designed and built a closed-loop, force-controlled fatigue rig to validate the fatigue life of deFlex's compliant mechanisms, capturing a full force-displacement curve each cycle at 100,000+ cycles/day for under \$1,000
- Used AI-assisted development to implement product features and rebuild the platform UI from scratch, streamlining the flexure design workflow

Engineering Intern - Infiltrator Water Technologies, Old Saybrook, CT

May 2025 – Aug 2025

- Led ops-automation to bring ~400,000 caps/year production in-house, improving margins by ~\$50K/year
- Evaluated robots + packaging redesign projecting 50–75% fewer manual interactions/shift
- Autonomously iterated through 26 designs using additive manufacturing and a Design-FEA-Test loop, de-risking tooling
- Rapidly learned DFM for injection-molding and released 3 components; added a clip feature that eliminated install time
- Developed a low-cost modification package for a high-level alarm by integrating a float valve into existing infrastructure
- Set up, tested and installed leading edge R&D projects; created 50+ drawings enabling supplier negotiations
- Drafted 8020 + vacuum-cup end-of-arm tooling for cap handling

ENGINEERING EXPERIENCE

Incoming EV Chief Engineer / Powertrain Lead – UConn Formula SAE

(Elected EV Chief for 2026-2027) May 2024 – Present

- Led the first and second electric vehicles to a 14th place and 5th place national finish respectively, owning the powertrain, accumulator, and electrical boxes
- Architected a 400-cell, 415 V, 8.32 kWh accumulator that passed technical inspection for the second year in a row
- Collected data to simulate multiple battery pack designs to create a decision matrix weighing cell selection and configuration
- Optimized accumulator heat paths and increased minimum busbar cross-sectional area to cut peak cell temperatures
- Led the design and manufacturing of the EV powertrain using Ansys for structural/topology optimization
- Designed water-proof electronics enclosures to house critical components
- Analyzed telemetry and tuned motor controller settings to fix drivability issues and reduce power cutbacks with Race Studio 3
- Coded a project-management web app with discord syncing, single-sign-on (SSO), and a timeline for critical path tracking

EV Powertrain Engineer – UConn Formula SAE

Aug 2023 – May 2024

- Reverse engineered a Yamaha R6 output shaft and developed a MATLAB simulation to determine optimal gear ratio
- Manufactured and assembled the EV powertrain in only a week using a mill, plasma cutter, and hydraulic press
- Analyzed powertrain using Ansys modal to shift modes of the structure above the motor's operating range
- Designed motor/differential mounts, and interfacing structure, using Ansys static Structural/Topology
- Redesigned the pedal mounting for 2-axis machining using Fusion 360 generative design cutting weight by 43%

Private Pilot

June 2021 – Present

SKILLS

CAD Software: SolidWorks, Creo, Fusion 360

Engineering Software: Ansys Workbench, LS-DYNA, Simulink, MITCalc, RMS GUI, Microsoft Office, Altium, VS Code, Race Studio 3

Programming Languages: Python, MATLAB

Languages: English (fluent), Polish (fluent)

Tools: Calipers, 3D Printers, CNC Mill, Plasma Cutter, Welder, Lathe, Spot Welder, Soldering, Band Saw, Drill Press